READER'S EDITION / VERSION 6.2

Unified Economic Simulator

Explained GPU Formulas

What the economy contains, what each equation means,
and how the same rules can execute in parallel.

This edition is written for someone encountering the project for the first time. It introduces the agents and institutions, distinguishes actual economic events from analytical measures, defines the notation, and explains the simplified formulas through small worked examples.

The central design is a population of distinct people, households, workplaces, firms, banks and institutions. Their actions require actual resources and valid agreements. A GPU evaluates many independent actions together while preserving the causal order of competing commitments.

What changed in this edition. The v6.1 formulas now have context: the question each answers, the meanings and units of its inputs, its assumptions, an interpretation of its result, and the operations needed to compute it. No new economic mechanism, numerical profile or calibration is introduced.

Status. This is a mathematical and execution specification. The attachments include earlier CPU reference checks and a reduced source implementation. They do not constitute a completed GPU engine, a hardware benchmark or evidence that a historical crisis has been reproduced.

Preserved material. The complete 15-page v6.1 manual and its inherited attachments, including the complete v6.0/v5.1 reference, remain embedded in this PDF. The main text can be read without opening them; they supply exhaustive source rules and provenance.

Choose a reading route.
For the economic idea, begin with Sections 1–3, then read the policy chapters. For the mathematics, read the notation primer and Sections 8–12. For implementation, read Sections 5–7 and 17. A first reading can skip the proof paragraphs while retaining the assumptions and examples.

Prepared for Kamran Orhun
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1 What this simulator is trying to represent

The project is a proposed agent-based economic simulator with Marxian analytical accounts. *Agent-based* means that the economy is built from distinct decision-makers and their relationships. A household has its own money and needs; a workplace has its own machines and workers; a bank has its own reserve account and borrowers. Aggregate output, wages, unemployment and crises are calculated from the history of these agents.

The aim is to study how production, exchange, social reproduction, ownership, credit, research and collective action interact over time. A shortage may stop a factory. A late payment may stop a purchase. Workers may refuse an offer or organize a strike. A government may authorize a service but still lack staff or funds to deliver it. These consequences must arise through the enabled rules and actual state, rather than through an imposed aggregate outcome.

This document explains the compact GPU formulas introduced in v6.1. It preserves their conditions. A simplified expression is useful only when it still answers the same economic question.

1.1 Who exists in the world?

Object	What it records	What it can do
Person	Identity, household, qualification, health, fatigue, available time, jobs and beliefs.	Work, care, learn, communicate, consume through a household, organize or refuse an activity.
Household	Members, money, usable necessities, housing/service access and unmet needs.	Pool permitted resources, seek provision, buy goods and evaluate whether members can afford to refuse work.
Workplace	Recipe, input stocks, equipment, jobs and unfinished production.	Perform dated production stages using actual materials, time, power and machine service.
Firm or corporation	Ownership/control links, accounts, contracts, financing and investment plans.	Operate one or several workplaces, quote prices, hire, borrow, invest and receive realized receipts.
Bank or monetary issuer	Deposits, reserves, loan claims, allowances, equity and payment queues.	Create or settle permitted claims, process payments and lend under the active institutional rules.
Organization	Membership, mandate, treasury, plans and support commitments.	Coordinate proposals, funded support and collective action; individual consent and feasibility still matter.
State/public institution	Jurisdiction, law, offices, budget, treasury, staff and implementation capacity.	Enact authorized policies, collect or spend funds, provide services and change rights at their declared effective times.
Research project	Required capabilities, access, attempts, resources and executed effort.	Attempt a discovery or imitation, record failures and successes, and enable later mastery or adoption.

A workplace is a physical productive unit; a corporation is an ownership and decision-making unit. Several

factories can belong to one corporation, and their ownership can differ from who has physical custody or effective control. A recipe is a description of inputs and outputs. It does not itself confer ownership, knowledge, permission or capacity.

The world also contains relationships: employment, ownership, loans, suppliers, communication contacts and research prerequisites. An agent can have many relationships even if its daily search or observation budget exposes only a few of them. Limiting observation is a behavioral assumption; deleting the unseen relationships would change the economy. [M1, S12–S13; M2, G1–G2]

1.2 Three layers that must stay distinct

Layer	Question it answers
Actual history	What goods existed, what work occurred, what was paid, what is still owed, and who controlled each resource?
Behavioral rules	What does an agent propose, given the information it is allowed to observe, its goals and its constraints?
Analytical accounts	What labor requirements, reproduction burdens, profit measures or other diagnostics describe a declared reference context or recorded history?

For example, a bakery can estimate that tomorrow's bread sales will cover wages. That forecast can influence its proposal, but it is not today's money. The bakery's actual bank balance decides whether a required current payment can settle. Separately, the analyst can calculate the bread's labor requirement from its recipe and upstream inputs. That labor value does not automatically become the bakery's posted price.

These distinctions allow expectations to be mistaken, goods to remain unsold, and a technologically feasible plan to fail for financial or institutional reasons.

2 The economic ideas behind the notation

2.1 Use, value, price and payment

The following meanings are used in this proposed model. They are not interchangeable units.

Term	Meaning in this model
Use-value / service	The practical contribution of a good or activity to a need or productive purpose. Food, care and power supply different services; their usefulness is not one universal monetary quantity.
Labor value	The direct and upstream labor requirement of reproducing an ordinary good under a specified reference technique and labor-type conversion. The vector v measures this in reference labor-hours per physical unit.
Market quote / price	The monetary terms proposed by an identifiable seller. Whether a transaction occurs depends on offers, consent, funds, stock, access and the settlement rules.
Labor power	The capacity to work, whose ongoing reproduction involves necessities, rest, care, training and social conditions. Its reference reproduction requirement and its observed monetary cost are different quantities.

Term	Meaning in this model
Wage contract	An agreement for specified work and gross pay, with applicable overtime, deductions and constraints. Actual received net pay is determined by performed or guaranteed work and settlement.
Claim	An enforceable or otherwise recognized obligation recorded between parties. A deposit, an unpaid wage and a loan are different instruments with different spending and settlement rules.
Monetary profit	A dated accounting outcome from recognized receipts/revenue, costs, expenses and the chosen inventory/valuation rules. It is not obtained merely by subtracting labor values.

A commodity can be useful and still fail to sell. A low reproduction labor requirement does not force a low market price when access is restricted. An unpaid wage can be recognized as an obligation while the household still lacks cash for a purchase. A profitable-looking borrower can face a liquidity shortage before its expected receipts arrive.

2.2 What is Marxian here, and what is a chosen modeling rule?

The framework explicitly tracks production and reproduction, ownership and control, labor power, surplus, accumulation, circulation, class organization and conflict. It also keeps the distinction between labor-value accounting and money accounts. These categories shape which state and causal mechanisms the simulator represents.

The numerical coefficients and behavioral policies are *modeling choices*. A particular bargaining weight, strike-participation sigmoid, research hazard or household adjustment rate is a declared hypothesis, not a theorem deduced from Marx. The optional system that derives labor-type conversion weights from reproduction and training is also a chosen analytical closure. Scenarios must identify their assumptions and parameters. [M2, E2, E9–E10, E13, E19]

In the inherited analytical account, selected productive living labor $L^{\dagger}$ can be compared with a declared reproduction or compensation requirement V_{χ}:

$$S_{\chi} = L^{\dagger} - V_{\chi}, \qquad e_{\chi} = \frac{S_{\chi}}{V_{\chi}} \quad \text{when } V_{\chi} > 0. \tag{2.1}$$

Here χ labels the comparator convention. All three quantities use compatible labor units and the same period. The classification of productive labor, the selected population and the reproduction bundle must be stated. These are *labeled diagnostics*; the ratio is not automatically a universal measurement of exploitation. A negative comparator does not by itself establish a negative observed capitalist exploitation rate.

The normative reproduction convention includes the designated worker households even when some members are unemployed or underemployed. A compensation-based comparator uses its declared affordable basket; that basket is not automatically what households actually consumed. Research effort, unpaid care and pure circulation/title transfers remain visible in their own relevant accounts. [M2, E2; E19.2]

A monetary profit rate additionally needs a monetary profit numerator and a consistent capital-advanced denominator. Unsold output can embody labor while producing no sale revenue. Rent and interest can transfer realized income without being new physical output. The model therefore keeps event history and the chosen analytical classifications available for separate reports.

2.3 Knowledge is reusable, but production still consumes resources

An invention creates a capability or piece of knowledge that can be used repeatedly. An ordinary product instance uses a recipe that may require that knowledge. This model records the historical effort spent on research separately from the labor needed to reproduce a later instance, conditional on knowledge already being available.

For software, for example, discovery or development effort belongs to the research history. Copying, distribution, operation or support may still use resources. The copying requirement can be very small without denying the effort that produced the software. A patent or license can limit access or create a payment claim; it does not, by itself, add the same past development labor to every copy's conditional reproduction requirement.

This is an accounting convention adopted for this simulator. The research formulas in Section 10 make the separation explicit. They also count failed attempts and avoid charging the same input labor twice.

3 Follow one simulated day

A day is divided into economic phases. A phase is an ordered part of simulated time, not a graphics frame or a GPU thread. Some phases have smaller subphases because events within the day affect what can happen next.

Step	Purpose	What becomes possible
0	Opening	Release previously completed output and contracted deliveries; process scheduled finance and authority changes in their prescribed order.
1	Observe	Expose only information agents may see, including declared lags. No one receives future outcomes.
2	Propose	Form quotes, applications, bounded bargaining, support and political proposals. A proposal changes no stock by itself.
3	Commit	Jointly reserve the actual resources required by accepted commitments: money, time, materials, equipment, support and collateral.
4	Work and pay	Deliver enabling care or power before dependent activity. Perform work and settle its required pay.
5	Provision and buy	Use actually received wages; deliver public services and grants; recompute shared household gaps; settle purchases in the prescribed sequence.
6	Physical close	Update health, fatigue, equipment and executed research; record output or hazards that become available later.
7	Accounts and decisions	Accrue obligations, tax, reconcile and approve funded decisions with future effective dates.
8	Audit	Check resources, claims, time and control, then checkpoint the full state and required history.

A wage can fund today's purchase only after it is received

Consider a teaching example using one currency and same-bank payments, with taxes omitted for this illustration. A household opens with 10 currency units, an employer with 20, and a shop with 0. The employer commits and pays 6 for two hours of actual work. The balances become 16, 14 and 0. Later, the household buys three already available units of food at 4 each, spending 12. The balances become 4, 14 and 12. Total deposits in these transfers remain 30. The shop gives up three food units; the household receives their goods or delivered service under the relevant consumption rule.
The opening balance of 10 could not fund this 12-unit purchase. The wage makes it possible only after settlement. If enabling childcare failed, the dependent work and pay would have to follow their failure rules. If the shop had no released food, money alone would not create it.

This order is economically consequential. A single snapshot for the whole day would miss the wage-funded purchase. Letting each GPU thread execute an entire agent's day independently could let some agents purchase before other agents' required wages or deliveries. The execution schedule must preserve these boundaries.

The source retains finite observation, offer and bargaining budgets where those profiles apply: five visible offers, two applications, two bargaining rounds and one active research project. The C6 banking profile has two scheduled payment-queue passes; the reduced P0 implementation has its separately specified convention. A mathematical iteration used to compute a labor diagnostic does not authorize unlimited market search or repeated banking settlement until equilibrium. [M1, S12; M2, G8, E17–E18]

4 How to read the mathematics

4.1 The small vocabulary of symbols

Notation	Read it as
x, x_i	One quantity; or the entry for object i in an array of quantities.
x' or x^+	The updated value after the named event or iteration. This is not a derivative.
$\widehat{x}$	An estimate, forecast or numerical candidate; the local paragraph says which. It is not automatically an observed fact.
$x \leftarrow y$	Replace the stored value of x with y.
$\sum_i x_i$	Add the entries over the stated index set.
min, max	Choose the smallest or largest permitted candidate. Their domains matter.
$[x]_+ = \max(0, x)$	The positive part: keep a positive amount, otherwise return zero.
$\lfloor x \rfloor$, $\lceil x \rceil$	Round down or up to an integer, respectively. A grid subscript instead means rounding to that grid.
$x \in \mathbb{R}_+^n$	An array of n nonnegative real quantities, usually with physical units. $\mathbb{Z}$ denotes integers.
A_{ij}	Entry in row i, column j of matrix A. Here a production column describes one output recipe.
A^{T}	The transpose: exchange rows and columns. The superscript is not simulated time.

Notation	Read it as
$v^{\mathsf{T}}y = \sum_i v_i y_i$	A dot product: pair each quantity with its corresponding weight, then add.
$A \geq 0$	Every entry is nonnegative. It does not compare determinants or eigenvalues.
$\cup$, $\setminus$	Union of sets, or remove members of the second set from the first.
$\rho(A)$	Spectral radius: the largest magnitude of any eigenvalue of A. Section 11 explains why it appears.

Indices often mean good i, output recipe j, occupation k, agent, date or event. A formula that updates one record may suppress these indices to stay legible. The enclosing section states the relevant record and time.

Symbols have local scope. The source reuses letters. In the value system, H is a labor-requirement matrix; in payroll, H is actual hours. In the value system, T is a training matrix; $T(y)$ is a tax function in the wage chapter, and T is an installment count in the loan formula. Likewise β is an occupational weight vector in one chapter and a bargaining scalar in another. Definitions beside each formula, rather than a single guessed global meaning, govern the interpretation.

4.2 Units are part of the formula

A production input coefficient has units such as kilograms of grain per loaf. Multiplying it by loaves to produce gives kilograms of grain. A labor value has units of reference labor-hours per loaf; multiplying it by a number of loaves gives reference labor-hours. A wage is currency per pay-hour. These quantities cannot be substituted for one another without an explicit conversion convention.

All equations that combine analytical quantities refer to one declared *reference context*: goods and units, selected techniques, labor types, classification, access assumptions, reproduction/training coefficients and their version. A context is a reproducible description, not an assertion that every actual factory uses the same technique or that every reference plan can execute today.

4.3 What makes an operation GPU compatible?

A GPU executes many instances of relatively regular work. In this design, an array can contain one health value per person, one stock quantity per inventory record or one edge per relationship. A *kernel* applies an operation to many such records.

Elementwise arithmetic updates each record; a *gather* reads selected neighboring records; a *reduction* sums or takes a minimum/maximum; a *prefix scan* calculates all running totals. A sparse matrix stores only represented nonzero relationships. A bitset encodes membership using bits, so set union can use bitwise OR.

These operations help express the model efficiently. They do not remove its resource or causality constraints. In particular, many firms can calculate proposals in parallel, but two proposals cannot both spend the same exclusive stock. Sections 5 and 17 explain the distinction.

5 Stocks, complete events and payments

5.1 Store amounts together with their units

For each resource or account coordinate r, store an integer n_r and declare a positive unit quantum ϵ_r:

$$\boxed{X_r = \epsilon_r n_r.} \tag{5.1}$$

X_r is the actual physical or monetary amount. If one money tick is 0.01 currency units, $n_r = 1234$ means 12.34. If one material tick is 0.001 kilograms, the same integer means 1.234 kilograms. A coordinate identifies the owner or custodian, resource or instrument, unit, and required attributes such as location or quality.

Spendable balances and physical stocks require nonnegative quantities. Some accounting coordinates and event changes are signed. Different currencies, goods, person-hours, machine services and emissions allowances cannot share an untyped balance.

Integer storage makes exact matching possible *within the declared units and proven ranges*. It does not make every fraction representable or make arithmetic overflow harmless. Check products and sums as well as final balances. Unsupported amounts need a declared finer scale, exact residual route or pre-execution rejection. Never wrap an integer, saturate an account, erase a residual or repair an invalid balance by clipping it to zero. [M1, S2; M2, G3]

5.2 An event is an indivisible economic commitment

A purchase is more than subtracting money: the payer must be eligible, the seller must control the goods, payment must be feasible, and delivery must occur under the contract. These linked changes form an *event bundle*. A *leg* is one signed change to one account or resource record.

Let $\mathcal{E}_{\text{ok}}$ be the events that have passed all required checks. Let δ_{re} be event e's signed integer change to coordinate r. Then

$$\boxed{n'_r = n_r + \sum_{e \in \mathcal{E}_{\text{ok}}} \delta_{re}.} \tag{5.2}$$

An accepted event contributes every required leg. A rejected event contributes none. This formula describes *posting accepted changes*; it does not itself decide which events are affordable or authorized.

Transfers conserve the relevant transferred instrument across their two sides. Production transforms goods, consumption destroys inventories or delivers services, and lawful issuance creates particular instruments. The common update therefore does not imply that every good, every money definition or every signed accounting field must have zero total change.

Reservation means holding a resource for a permitted commitment so it cannot also be promised elsewhere. It is not automatically a payment or consumption event. At commitment and settlement, the model checks required gross resources together, including any shared constraints. A worker has one time budget across employers, a bank has one actual reserve account, and support programs share the same household need gap.

A rejected buyer must leave the item for the next buyer

A seller has one item priced at 3. Buyer A has 2 and has priority over buyer B, who has 3. Assume delivery, rights and all other conditions are valid.
A's complete purchase fails the funding check. No money or goods change hands. B is then checked against the still-available item and succeeds. The seller receives 3, B receives the item, and A keeps 2.

A rejected buyer must leave the item for the next buyer (continued)

A parallel routine that first allocates the item to A, then rejects A's payment without reconsidering B, produces the wrong history. Resource allocation and payment feasibility must belong to the same ordered bundle resolver.

5.3 Why bank reserves appear in a payment

A *deposit* is a customer's claim on its bank. A *reserve balance* is the bank's settlement asset with the applicable monetary issuer. They are different instruments. When customer p at bank a pays customer q at bank b, the ordinary cross-bank transfer of s has four legs:

$$(\Delta D_p, \Delta R_a, \Delta R_b, \Delta D_q) = s(-1, -1, +1, +1). \tag{5.3}$$

D_p, D_q are the customer deposits; R_a, R_b are bank reserves, in compatible units for the transfer. The payer's deposit falls, its bank transfers reserves, the receiving bank gains reserves, and the payee's deposit rises. Customer records and the bank's liability view refer to the *same* deposit claims; they are not two separate spendable assets.

A five-unit cross-bank payment

Before payment, let $(D_p, R_a, R_b, D_q) = (12, 7, 3, 1)$. Paying $s = 5$ gives $(7, 2, 8, 6)$. Across these entries, customer deposits total 13 before and after; bank reserves total 10 before and after.
If the paying bank had only 4 unreserved reserve units, the same payment would fail or queue under the active protocol even though the customer has 12 on deposit. A separately authorized, actually funded liquidity facility can change that state; the customer's deposit balance alone cannot.

For a same-bank payment, only the two customer deposit legs are needed. Loan creation, repayment, impairment and reserve issuance have their own matched event templates. They must not be replaced by this transfer formula.

This is the basis for distinguishing *liquidity*, the ability to meet due payments using currently available settlement assets, from *solvency*, the relationship between assets, liabilities and equity under the valuation rules. A bank can face a payment shortage without the simulator inventing new equity; a loss in asset value is not itself the legal discharge of the borrower's debt. [M1, S2, S12; M2, G14, E5, E17]

6 Exact production packets and fractional obligations

6.1 Compile a recipe into a representable activity unit

The question is: how can a fixed recipe consume the correct proportions when the ledger uses integer ticks?

Consider one dated activity stage. A leg ℓ has physical coefficient c_ℓ, resource quantum $\epsilon_{r(\ell)}$, and the proposed activity unit is ϵ_a. Its ticks per activity unit are

$$\theta_\ell = \frac{c_\ell \epsilon_a}{\epsilon_{r(\ell)}} = \frac{p_\ell}{d_\ell}, \tag{6.1}$$

with the fraction reduced and $d_\ell > 0$. Include output, materials, work, machine service and any immediate linear invoice that belongs to this stage. A packet contains the smallest integer number of these proposed

units that makes every leg integral:

$$\boxed{k = \operatorname{lcm}_\ell d_\ell, \qquad b_\ell = k\theta_\ell \in \mathbb{Z}_{\geq 0}.} \tag{6.2}$$

Here lcm is the least common multiple. A packet represents $k\epsilon_a$ activity units. Keep gross input ticks b_r^- and output ticks b_r^+ separate, even when a process consumes and returns the same material.

One packet preserves all recipe ratios

For this teaching example, one unit of finished output requires 1/2 material unit and 1/3 work-hour. All three resource quanta are one unit. The denominators are 2, 3 and 1, so $k = 6$.
One packet therefore uses 3 material units and 2 hours to deliver 6 output units at the stage's permitted release time. There is no rounding away of either input. The chosen coarse quanta force six-unit packets; finer declared quanta could permit a finer activity lattice.

Given *exclusively assigned* available resources n_r^{avail} and a nonnegative integer plan q_{plan}, the possible packet count is

$$q = \min\left(q_{\text{plan}}, \min_{r:b_r^->0} \left\lfloor \frac{n_r^{\text{avail}}}{b_r^-} \right\rfloor\right), \qquad \Delta n = q(b^+ - b^-). \tag{6.3}$$

The inner ratio asks how many packets each resource can support. The smallest is the binding resource. An empty inner minimum is $+\infty$; other gates can still limit the activity. Apply Δn only after the complete bundle is accepted.

For the packet above, a plan of four packets, eight material units and five hours yields

$$q = \min(4, \lfloor 8/3 \rfloor, \lfloor 5/2 \rfloor) = 2.$$

It uses six material units and four hours for twelve output units. Other constraints, such as pay funding and access, are assumed satisfied in this illustration.

Gross inputs matter. A process that consumes three units and returns two must have three initially; its net use of one is not enough for the reservation. Group all simultaneously required debits to the same resource before checking.

Timing matters. A one-time start input, a later work slice and a delayed release remain different dated stages. Packet compilation cannot charge future work at the start or deliver future output early. Tiered taxes, overtime and changing prices are not one globally linear coefficient; evaluate their active branch and revalidate the event. Recompile when recipe, units or defining assumptions change.

Shared inputs matter. Two jobs cannot each treat the same inventory as exclusively assigned. Their packet bounds are proposals until the ordered resolver grants their competing commitments. Packet size and integer range must be checked in advance; increasing a packet can change feasible actions. [M1, S3]

6.2 Keep a fractional obligation until it is settled

A cash unit may be coarser than an earned obligation. For one debtor, creditor, currency and fixed-denominator tranche, store one nonnegative integer numerator:

$$N = dD + r, \qquad \text{face amount} = \frac{N}{d}, \qquad d > 0, \quad 0 \leq r < d. \tag{6.4}$$

D is the number of whole cash quanta owed; r is the fractional remainder. The denominator d belongs to this tranche. A tranche is a separately tracked portion of a contract with compatible accrual rules.

If a numerator units are earned per accrual unit and k such units occur, then

$$N \leftarrow N + ak, \tag{6.5}$$

$$N \leftarrow N - ds, \qquad 0 \le s \le \lfloor N/d \rfloor \qquad \text{when an actual cash payment } s \text{ settles.} \tag{6.6}$$

The payment also needs actual funds and all applicable contract checks. Its cash legs are $-s, +s$. A larger N records a larger obligation; it does not automatically credit the recipient's deposit account.

Three thirds are kept, not discarded

Start with $N = 0$ and $d = 3$. Three separate earned increments of one-third of a cash quantum give numerators $1, 2, 3$. Before the third increment, the remaining fractional claim is still recorded. Once one whole cash quantum is payable and funded, paying $s = 1$ gives $N = 3 - 3 = 0$.
If the payer has no cash, N remains 3. Recognition of the debt and ability to buy groceries are different facts.

Different counterparties and currencies do not share a remainder bucket. Different denominators need exact conversion or separate tranches. Compounding can require a new denominator; it is not automatically representable by the old one. Check the widths of N, ak and ds.

The full face amount enters the accounts that the selected profile recognizes. In particular, the reduced P0 source's memorandum-interest convention is not silently converted into another profile's accrual asset. Contract closure does not erase a nonzero obligation. Net pay, withholding and gross pay partition the same obligation, and deferred fractions require contractual permission. Current-cash wage guarantees still need actually settled, representable payments. [M1, S4]

7 Reusable policy operations and reproducible draws

7.1 Bounds, grids and smoothing

Many policies use the same small mathematical operations:

$$[x]_+ = \max(0, x), \qquad \operatorname{clip}(x; l, u) = \min(u, \max(l, x)), \qquad l \le u. \tag{7.1}$$

The positive part can express an unmet need or funding gap. Clipping expresses an economic bound, such as a health index between zero and one. It must not be used to hide accounting errors or numerical overflow.

Prices and wages may live on a *grid*: permitted increments of size $\epsilon > 0$. For a quote with legal floor l and cap u, define

$$L = \lceil l/\epsilon \rceil, \quad U = \lfloor u/\epsilon \rfloor, \quad \operatorname{Quote}(x) = \epsilon \operatorname{clip}(\lceil x/\epsilon \rceil; L, U). \tag{7.2}$$

This chooses an upward-rounded quote on the legal grid, provided $L \le U$. If the interval contains no legal grid point, the offer is unavailable. For an illustrative one-cent grid, candidate 2.001 and cap 2.009 produce a legal quote of 2.00 if that value satisfies the floor: the cap is first converted to its largest legal grid point. With floor 2.005 and cap 2.009, the legal interval is empty.

This is the source's product-quote rule. Wage stages keep their own specified upward, downward or nearest rounding. Exchanging those directions can change whether a contract meets a worker's required pay.

Forecasts and beliefs often use the affine blend

$$\mathrm{L}(x, y; \lambda) = x + \lambda(y - x), \qquad 0 \le \lambda \le 1. \tag{7.3}$$

x is the previous belief, y is the admitted new signal, and λ is the update weight. At $\lambda = 0$ nothing changes; at $\lambda = 1$ the signal replaces the old value. For $x = 10$, $y = 14$ and $\lambda = 1/4$, the result is 11.

The algebra equals $(1 - \lambda)x + \lambda y$, but moving a specified intermediate rounding can change the numerical answer. Preserve the original rounding and the signal's permitted observation window. A missing observation takes its missing-data branch; it is not a measured zero. [M1, S5.1]

7.2 Nonlinear functions retain their specified shapes

Policies can use exponentials, powers or a sigmoid, such as

$$h(z) = 1 - \exp(-z),\ z \geq 0, \qquad \text{sigmoid}(z) = \frac{1}{1 + \exp(-z)}.$$

A hazard converts nonnegative exposure or effort into a probability-like response. A sigmoid maps a score to a number between zero and one. The relevant economic chapter defines the score and what the resulting probability controls.

GPU-D1 evaluates the declared policy functions with versioned integer tables, checked domains, prescribed interpolation and rounding. Such a table is a pinned approximation introduced by that execution profile. This edition does not replace the curves with straight lines or change their thresholds. If a calculation is outside the supported domain, the runtime reports or recovers from that numerical condition; it does not invent an economic outcome.

Tax brackets, consent, eligibility, legal authority and other discontinuous decisions remain explicit branches. A shared implementation function is legitimate only if it preserves each policy's declared numerical result. [M1, S5.2, S9]

7.3 Randomness is keyed to economic identity

A seed allows a stochastic model to be replayed only if the same logical event receives the same random draw. The source keys SHA-256 draws by seed, day, logical economic domain, persistent entity or event ID, and draw index. A persistent ID survives sorting and memory compaction. A draw must not depend on a GPU thread number, memory slot, completion order or number of scheduling waves.

Let u_e be the source-prescribed 128-bit draw, interpreted as an integer from 0 through $2^{128} - 1$. For probability a/b with integers $b > 0$ and $0 \leq a \leq b$, the retained threshold rule is

$$T_{\text{draw}} = \left\lfloor 2^{128} a/b \right\rfloor, \qquad \boxed{\text{success if } u_e < T_{\text{draw}}.} \tag{7.4}$$

T_{draw} is named explicitly here to distinguish it from a training matrix or tax function. Probability one needs a separate tag because the threshold 2^{128} lies outside an unsigned 128-bit draw's range. Zero effort has a zero success threshold where the source requires it; finite positive hazards keep their stated cap below probability one.

The floor is part of the rule. In a toy two-bit analogue with four possible draws and probability 1/3, the threshold is $\lfloor 4/3 \rfloor = 1$, so only draw 0 succeeds. Replacing it with $3u_e < 4$ also accepts draw 1. That apparently simpler comparison would change the model.

Computing this rule exactly on a chosen GPU backend may need multiword integer operations or a declared fallback; writing a short equation does not prove the backend has a native 128-bit instruction. Cache an unchanged threshold where permitted. [M1, S5.2]

Three kinds of numerical statement. An algebraic identity is a claim about mathematics. Exact integer accounting is a claim about representable units and proven bounds. Bitwise replay is a claim about the entire implemented transition pipeline, including policies, reductions, ordering and versions. None alone proves the other two.

8 Goods and labor values

The production account answers a concrete question: given a selected way of making a good, how much labor is required both at the final workplace and throughout its input chain? A bakery uses grain, and producing that grain also uses labor and inputs. Counting only hours inside the bakery would omit this upstream work. The equations below calculate a *conditional reproduction requirement*: a requirement under a stated production context. [M1, S6; M2, G9]

A **context** identifies the goods, their physical units and qualities, the selected production method for each output, the labor categories, and the date/version to which these choices apply. One column describes one selected method. Two alternative methods are alternative selections, not two sets of inputs that must both be consumed. A discovery can make a method available; adoption changes the selected context when the operational conditions actually change.

Reference labor requirements, performed hours, wages, cash prices, and usefulness are different records. A value calculation does not create money, move inventory, pay a worker, or establish that a plan can start today. Revaluing an existing stock does not count as producing it again.

8.1 Read a recipe column before reading the matrix

There are n ordinary goods and m occupations. All vectors below are column vectors. A superscript T means transpose: exchange rows and columns. Entries have declared units, even when the compact notation suppresses them.

Symbol	Shape	Meaning of an entry
A	$n \times n$	A_{ij}: units of input good i per unit of output good j. Column j is its material recipe.
H	$m \times n$	H_{kj}: direct clock-hours of occupation k per unit of output j.
α	m	α_k: common labor units per clock-hour of occupation k. One common unit is the chosen reference occupation's hour. These are conversion weights, not wage rates.
v	n	v_j: common labor units required per physical unit of output j.
x, y	n each	Gross ordinary production x and final ordinary output y, in each good's own physical unit.

For one output j, add the labor requirements of its material inputs and its direct work:

$$v_j = \sum_{i=1}^{n} A_{ij} v_i + \sum_{k=1}^{m} H_{kj} \alpha_k. \tag{8.1}$$

An input coefficient has units of input per output; multiplying by labor per input gives labor per output. Direct occupation-hours multiplied by their conversion weight give the same unit. Therefore the two terms

can be added. The compact form is

$$\boxed{v = A^{\mathsf{T}} v + H^{\mathsf{T}} \alpha.} \tag{8.2}$$

The transpose appears because each output's recipe is a *column*. In contrast, physical intermediate demand for a production vector x is Ax: each row adds demand for one input across all output activities. Without research, gross requirements satisfy $x = y + Ax$.

8.2 One reusable solve, with a stated domain

For a nonnegative matrix $K \geq 0$ and a nonnegative right-hand side $b \geq 0$, write

$$q = \mathrm{Solve}(K, b) \quad \Longleftrightarrow \quad q = b + Kq \quad \Longleftrightarrow \quad (I - K)q = b. \tag{8.3}$$

Here I is the identity matrix, which leaves a vector unchanged. The intended nonnegative solution requires $\rho(K) < 1$; Section 11 explains this condition and numerical error. The symbol b in this operation is a vector, unrelated to the reference-occupation index used below.

The notation names a calculation, not an implementation that constructs $(I - K)^{-1}$. A sparse implementation stores nonzero coefficients and their indices, then applies weighted sums only along the corresponding relationships. An iterative method may repeatedly calculate $q^{+} = b + Kq$ until its error is bounded. Another verified linear solver may compute the same answer. This analytical fixed point is not an instruction to clear markets, prices, or payment queues until equilibrium.

For externally specified positive weights, the required goods calculation is simply

$$\boxed{v = \mathrm{Solve}(A^{\mathsf{T}}, H^{\mathsf{T}} \alpha).} \tag{8.4}$$

This reference layer assumes a finite selection of single-output, constant-coefficient methods. Actual joint production, indivisible jobs, dated stages, and nonlinear capacities retain their physical event rules. A finite value vector does not prove the availability of seed stock, machines, workers, funding, or permissions.

9 Optional occupational conversion

Actual work is first recorded in occupational clock-hours. To report a single scalar labor measure, the model needs a convention for converting those hours. It can take positive weights α as supplied inputs, including equal clock-hour weights. Alternatively, it can use the following *optional reproduction-based closure*. This is a declared modeling assumption, not a wage equation or a universal result about skilled work. [M1, S6.1–S6.2; M2, G9 and Foundation 13]

9.1 What reproducing an occupation means here

For each target occupation j, S_{ij} is recurrent reproduction goods of type i per productive hour; E_{ij} is education-related goods per newly qualified worker; and U_{kj} is additional direct training hours of teaching occupation k per newly qualified worker of type j. The finite number $\gamma_j > 0$ is that worker's productive lifetime hours under the selected stationary-cohort or planning convention.

Convert these to requirements per supplied occupation-hour:

$$\boxed{P_{ij} = S_{ij} + \frac{E_{ij}}{\gamma_j}, \qquad T_{kj} = \frac{U_{kj}}{\gamma_j}.} \tag{9.1}$$

Thus P has shape $n \times m$ and measures goods per occupation-hour; T has shape $m \times m$ and measures additional training hours per occupation-hour. This local matrix T is unrelated to tax or elapsed time.

Labor already embodied in purchased education services in E must not also appear in U. Training-period subsistence follows the source convention and is not charged again in recurrent S.

The lifetime allocation of individual training is separate from the treatment of historical invention. It does not allocate an old research project's total labor to every later product copy.

9.2 Two equations, one coupled sparse calculation

Let u be the unnormalized goods-value vector and β the unnormalized occupational-weight vector. Here β is unrelated to bargaining power. The closure is

$$\boxed{\begin{aligned} u &= A^{\mathsf{T}}u + H^{\mathsf{T}}\beta, \\ \beta &= \mathbf{1} + P^{\mathsf{T}}u + T^{\mathsf{T}}\beta. \end{aligned}} \tag{9.2}$$

The first equation is ordinary labor accounting. The second adds the goods and training requirements of supplying an occupation-hour to a common residual, initially set to one for every occupation. The vector $\mathbf{1}$ contains m ones; the goods block's corresponding source vector contains n zeros.

Choose reference occupation b and normalize both vectors together:

$$\boxed{v = \frac{u}{\beta_b}, \qquad \alpha = \frac{\beta}{\beta_b}, \qquad \kappa = \frac{1}{\beta_b}.} \tag{9.3}$$

Then $\alpha_b = 1$, and the normalized closure is $\alpha = \kappa\mathbf{1} + P^{\mathsf{T}}v + T^{\mathsf{T}}\alpha$. This displays the assumption clearly: the same residual κ is assigned to each clock-hour after its selected reproduction burden. Neither κ nor β measures actual surplus labor. The weights are not human worth, observed wage ratios, or independently measured productivity ratios. Keep the underlying hours and inputs available for comparison with other conversion conventions.

The two equations can be passed to the common solver as

$$\begin{pmatrix} u \\ \beta \end{pmatrix} = \operatorname{Solve}\left(K, \begin{pmatrix} 0 \\ \mathbf{1} \end{pmatrix}\right), \qquad K = \begin{pmatrix} A^{\mathsf{T}} & H^{\mathsf{T}} \\ P^{\mathsf{T}} & T^{\mathsf{T}} \end{pmatrix}. \tag{9.4}$$

An implementation need not assemble this block matrix. It can apply four sparse products directly. The documented simultaneous iteration reads the same old u, β for both updates and writes separate new vectors.

9.3 What the reduction preserves

For comparison with the earlier formulas, define the full occupational-requirement matrix $G = H(I - A)^{-1}$ and the occupational-reproduction matrix $M = GP + T$. These are explanatory intermediates; the runtime need not construct them. The first block gives $u = G^{\mathsf{T}}\beta$. Substituting it into the second yields $\beta = \mathbf{1} + M^{\mathsf{T}}\beta$, the earlier closure. For these finite nonnegative blocks,

$$\rho(K) < 1 \quad \Longleftrightarrow \quad \rho(A) < 1 \text{ and } \rho(M) < 1. \tag{9.5}$$

Both admissibility requirements remain. If only the optional occupational closure fails, an otherwise admissible goods context may still use separately declared positive weights. Removing intermediate matrices reduces required storage; it does not guarantee faster execution. The coupled iteration can converge more slowly than a reduced solve, so solver choice remains an implementation and benchmark decision.

10 Research and a worked production example

Research consumes resources in an attempt to obtain reusable knowledge. An attempt's resource requirement and its outcome are separate: unsuccessful work still uses time and materials, while success updates the relevant discovery or mastery state at its prescribed date. Legal permission remains a separate condition. A research result is not consumed anew each time an available production recipe uses it. [M1, S6.3; M2, G9 and Foundation 8–9]

10.1 Requirements per attempt and per plan

There are r research-attempt types. The matrix $J \in \mathbb{R}_+^{n\times r}$ gives ordinary goods per attempt, and $F \in \mathbb{R}_+^{m\times r}$ gives additional direct occupational hours per attempt. Direct research work in F must be disjoint from labor embodied in the ordinary inputs in J. With v and α already calculated in the same context,

$$\boxed{\eta = J^{\mathsf{T}} v + F^{\mathsf{T}} \alpha.} \tag{10.1}$$

The vector η has r entries, measured in common labor units per attempt. It is not a cost per guaranteed discovery.

Let z be selected attempt counts and y final ordinary output. Counts obey each attempt catalogue's indivisibility rules. If the research inputs are freshly reproduced and are not also included in y, then

$$\boxed{\mathcal{V}(y, z) = v^{\mathsf{T}} y + \eta^{\mathsf{T}} z.} \tag{10.2}$$

A dot product multiplies each requirement by its quantity and sums the results. Once v and η are cached for the context, these two dot products give a scalar plan total. If gross output and occupational totals are also needed, calculate

$$\boxed{x = \mathrm{Solve}(A, y + Jz), \qquad \mathcal{L} = Hx + Fz.} \tag{10.3}$$

The vector $\mathcal{L}$ retains separate occupational clock-hours, and $\mathcal{V} = \alpha^{\mathsf{T}}\mathcal{L}$. The identity follows from $v^{\mathsf{T}}(I - A) = \alpha^{\mathsf{T}} H$:

$$\alpha^{\mathsf{T}} Hx + \alpha^{\mathsf{T}} Fz = v^{\mathsf{T}}(y + Jz) + \alpha^{\mathsf{T}} Fz = v^{\mathsf{T}} y + \eta^{\mathsf{T}} z. \tag{10.4}$$

This is an accounting identity under the selected conditions, not a physical-capacity or funding certificate.

Grain, bread, and an experimental recipe

The numbers are illustrative, not calibrated. There is one occupation, so $\alpha = 1$. Units are a grain unit, a loaf, and a clock-hour.

One unit of output	Grain input	Bread input	Direct hours
Grain	1/2	0	1
Bread	1	0	1

Thus $A = \begin{pmatrix} 1/2 & 1 \\ 0 & 0 \end{pmatrix}$ and $H = (1 \quad 1)$. The grain equation is $v_g = 1 + v_g/2$, so $v_g = 2$. The bread equation is $v_b = 1 + v_g = 3$. A loaf therefore requires one direct baking hour and two upstream hours in grain. Here $\rho(A) = 1/2$.

To illustrate the optional closure, take $P = (1/8 \quad 1/12)^{\mathsf{T}}$ and $T = (1/4)$. These are already-compiled per-hour reproduction/training coefficients. Then $GP = 2/8 + 3/12 = 1/2$ and $M = 3/4$. Hence $\beta = 1 + (3/4)\beta = 4$ and $u = (8, 12)^{\mathsf{T}}$. Normalization returns $\alpha = 1$ and $v = (2, 3)^{\mathsf{T}}$. With one

Grain, bread, and an experimental recipe (continued)

occupation, normalization removes its raw scale; this example does not demonstrate changing ratios between occupations.

Continue the same example: one research attempt

An experimental recipe attempt uses half a grain unit, one loaf, and two additional direct hours:

$$J = \begin{pmatrix} 1/2 \\ 1 \end{pmatrix}, \quad F = (2), \qquad \eta = (1/2) \cdot 2 + 1 \cdot 3 + 2 = 6.$$

Request two final loaves and one attempt: $y = (0, 2)^{\mathsf{T}}$ and $z = 1$. Ordinary production must cover $y + Jz = (1/2, 3)^{\mathsf{T}}$. Therefore

$$x_b = 3, \qquad x_g = \tfrac{1}{2}x_g + 3 + \tfrac{1}{2} = 7, \qquad \mathcal{L} = 7 + 3 + 2 = 12.$$

The compact total agrees: $\mathcal{V} = 3 \cdot 2 + 6 \cdot 1 = 12$ hours. Three gross loaves include two final loaves and one research input. Seven gross grain units cover 3.5 units of seed, three for baking, and half a unit for research. Gross production exceeds final delivery because it includes intermediate use.
These are requirement calculations. A dated execution also needs initial seed stock, workers, equipment, permissions, and funding. For example, an opening seed buffer of 3.5 grain units permits this grain batch and is restored after the indicated withdrawals; the matrix equation does not create that buffer.

10.2 Historical research is counted once

If a past research effort cost six hours and the bread recipe remains unchanged, later loaves still have a conditional requirement of three hours each. The six hours remain in the historical research account; the model does not add $6/N$ to each of N later copies. If adoption changes the recipe, calculate the new requirement in the new context. A project cannot use an input-production method that becomes available only through its own future success.

The same distinction applies to software: an absent release makes its copying route unavailable, not a zero-cost available route. Once accessible, copies or hosted services use their declared ongoing resources; historical development remains a separate record.

Stored inputs require chronological accounting. If producing an input took one hour yesterday and research consumes it with two more hours today, actual work totals three hours. Adding yesterday's hour to the research's fresh-reproduction requirement $(1 + 2)$ would incorrectly report four. Multi-period attempts likewise record each actual work stage once; completion does not charge the whole attempt again. When coefficients change between stages, use their dated contexts and retain raw occupational hours rather than applying one current matrix to the entire history.

11 Solver accuracy and unavailable results

An equation can be well defined while a particular numerical attempt has not yet solved it accurately. These are different questions. For a finite nonnegative operator K, its **spectral radius** $\rho(K)$ is the largest absolute value among its eigenvalues. An eigenvalue describes how the operator scales a corresponding direction. The condition $\rho(K) < 1$ makes repeated upstream requirements decay sufficiently for the

intended finite nonnegative solution of $q = b + Kq$. [M1, S7; M2, G10]

A production cycle is not automatically inadmissible: the grain recipe uses seed while producing a positive net output. Nor is every row sum required to be less than one. Conversely, running a fixed number of iterations supplies no accuracy guarantee.

11.1 Turn equation mismatch into an error bound

Choose a strictly positive scale vector w and a number $0 \le c < 1$ and certify

$$Kw \le cw. \tag{11.1}$$

The inequality is component by component. It establishes contraction under the weighted maximum norm

$$\|a\|_{\infty,w} = \max_i \frac{|a_i|}{w_i}. \tag{11.2}$$

The weights specify how coordinate errors are scaled. For a candidate $\widehat{q}$, its residual is the amount by which it fails its own equation:

$$r = b + K\widehat{q} - \widehat{q}, \qquad \boxed{\|q - \widehat{q}\|_{\infty,w} \le \frac{\|r\|_{\infty,w}}{1-c}.} \tag{11.3}$$

To see why, subtract the candidate from the exact equation: $q - \widehat{q} = K(q - \widehat{q}) + r$. The norm of the first term is at most c times the unknown error; moving that bound to the left yields the result. Stop only when the bound meets the declared tolerance.

For the grain/bread value equation, $K = A^{\mathsf{T}}$ and $w = (1, 2)^{\mathsf{T}}$ give $Kw = (1/2, 1)^{\mathsf{T}} = (1/2)w$. Thus $c = 1/2$. The candidate $(1.9, 2.9)^{\mathsf{T}}$ has residual $(0.05, 0)^{\mathsf{T}}$ and weighted error bound 0.1. Its individual coordinate errors are bounded by $0.1w$, namely $(0.1, 0.2)$. This is not the same as claiming a 0.1 unscaled error bound for every coordinate.

11.2 What a certificate must cover

Every actual operator needs its own certificate. A bound for A is not automatically a bound for A^{T} or for the coupled K. Near $c = 1$, the denominator amplifies residual error. Check the inequalities exactly or through justified outward error enclosures, including uncertainty in coefficients; an ordinary floating-point comparison is not by itself a certified bound.

In the coupled model, $\beta_b \ge 1$ exactly under the stated assumptions. Normalization still needs propagated error. For nonnegative bounds $u^- \le u \le u^+$ and positive $\beta_b^- \le \beta_b \le \beta_b^+$,

$$\frac{u^-}{\beta_b^+} \le v \le \frac{u^+}{\beta_b^-}. \tag{11.4}$$

Apply the corresponding positive interval division to each occupational weight as well. A small raw-solve residual does not automatically bound every normalized output.

Failure to find a certificate is not proof of nonproductivity. Exhausting a computation budget means numerical nonconvergence, not an economic crisis. Missing data, an inapplicable reference structure, demonstrated inadmissibility, and unresolved numerical accuracy require distinct status tags. None should be serialized as zero value or used to stop otherwise feasible physical history.

12 Knowledge, access, and executable readiness

Numerical recipes state how many resources are required; knowledge dependencies state which capabilities must exist along the chain. Capability requirements are logical conditions, not an extra quantity of historical research labor automatically consumed with each unit. [M1, S8; M2, G11]

12.1 Capabilities required somewhere in a production chain

For output j, let D_j be its directly required capability set and $\mathrm{pre}(j)$ its selected upstream goods. Let C_j collect capabilities required anywhere in its chain. Initialize $C_j = D_j$ and update simultaneously:

$$\boxed{C_j^+ = C_j \cup \bigcup_{i \in \mathrm{pre}(j)} C_i.} \tag{12.1}$$

The symbol $\cup$ means set union: include each item once. Repeat until no set changes. With n goods, at most $n-1$ propagation rounds after initialization suffice. Unlike the numerical solve, this finite logical closure needs no spectral-radius test. A GPU can gather neighbor sets and combine bitsets with bitwise OR, or merge sorted sets.

For example, grain might directly require agronomy and bread might require baking. Bread's chain support is then {agronomy, baking}. A bakery buying grain need not personally master agronomy: its grain supplier must satisfy that local requirement. Chain support documents what is needed somewhere in the selected chain, not a requirement that every downstream actor own all upstream knowledge.

Dependencies include all nonzero recipe-input edges with their declared engineering meaning. Deleted edges or changed access can require recomputation: an accumulating union cannot remove obsolete entries by itself.

12.2 Which research results are still missing?

Use $\mathcal{T}$ for target results and $\mathcal{A}$ for results already satisfied and accessible in the relevant actor/context. These are the sets called T, A in the compact manual; the script letters distinguish them from the training and input matrices. Start with $W = \mathcal{T} \setminus \mathcal{A}$ and repeat

$$\boxed{W^+ = W \cup (\mathrm{pre}(W) \setminus \mathcal{A}).} \tag{12.2}$$

Here set difference $\setminus$ removes already satisfied items, and $\mathrm{pre}(W)$ collects prerequisites of results in W. Satisfied accessible results stop traversal. A shared missing prerequisite appears once in this selected plan; multiple historical attempts remain distinct events. Actor and access context belong in the identity when satisfaction differs between producers.

A target needing both metallurgy and control software may have basic measurement as a shared prerequisite. If measurement is already accessible, no new measurement project is required. If it is missing, add it once. If two wholly missing results require one another, closing their set does not produce an executable first step: the scenario needs a seed, explicit bootstrap, or jointly specified activity.

> Keep discovery, mastery, permission, and adoption separate. A result may exist without this actor knowing how to use it; mastery may exist without legal or practical access; access may exist before equipment and a production method are installed. An activity starts only when its local prerequisites, actual resources, time, authority, and permissions are valid at its execution boundary.

Publication and communication occur at their prescribed phases. No capability closure changes an earlier observation or supplies missing funds, machines, or worker hours. This separation lets research alter future production possibilities while the event ledger continues to record what actually happened, once.

13 Households: needs, time and the capacity to work

Production requires people whose everyday needs can be met. A household owns provisions, holds money, cares for its members and purchases or performs services. A person has health, fatigue, qualifications, preferences and one time budget. A food purchase, for example, need not deliver a cooked meal if the chosen activity also requires fuel and cooking time. The formulas below connect actual provisioning to later work capacity. Their numerical coefficients are explicit behavioral assumptions, not deductions from the labor-value equations. [M1, S9.1; M2, E3]

13.1 Measure unmet needs before updating health

For each positive service need, compare the quantity required with the quantity actually delivered:

$$d = \sum_{j:n_j>0} \omega_j \frac{[n_j - s_j]_+}{n_j}. \tag{S9.1}$$

Here n_j is the required quantity of service j, $s_j \geq 0$ is its delivered usable quantity in the same units, and $\omega_j \geq 0$ is its aggregation weight. The weights sum to one over the positive needs. Thus $d \in [0,1]$: zero means every positively weighted need is satisfied, and one means every positively weighted need is wholly unmet. Zero-weight services remain separately recorded. If there are no positive needs, define $d = 0$. An unavailable required input is reported separately; it is not silently treated as a measured zero need.

A household deficit is a service shortfall

Suppose food service requires 2 units and receives 1, while housing service requires and receives 1. With equal weights, $d = \frac{1}{2}(2-1)/2 + \frac{1}{2}(1-1)/1 = 0.25$. Holding additional uncooked food does not remove a meal deficit until the recipe's other requirements are met. These numbers are a teaching example, not scenario defaults.

The weights summarize unlike services only after each shortfall has been divided by its own need. They do not add kilograms, housing-days and care-hours as if those were the same physical unit. The original individual service quantities remain in the state.

13.2 Every activity uses the same person's clock

The daily limit on paid work is

$$T = \min\left\{\bar{h}\, a(1-F),\ [24 - r_0 - c - e - v - o]_+\right\}. \tag{S9.2}$$

Local symbol	Meaning and unit
$T, \bar{h}$	Available paid-work hours/day; healthy-work cap after personal, legal and collective limits.
a, F	Health and fatigue, each in $[0,1]$.
r_0	Reserved rest allowance, in hours/day.
c, e, v, o	Reserved care, education, travel and organization hours/day.

The number 24 represents hours in one model day. The first bound represents health and fatigue constraints; the second prevents paid work from using time already needed elsewhere. Set $T = 0$ for a nonparticipant. Availability is not consent, and the cap is shared across employers. Delivered childcare can release time only in its actual delivery subphase. A promised service cannot release hours early. [M2, E3.1; E16]

13.3 Close the physical day using executed activities

After provisioning and work, update health and fatigue:

$$a' = \mathrm{clip}(a + \rho(1-a)(1-d) - \mu d - \nu F; 0, 1)\,, \tag{S9.3}$$

$$F' = \mathrm{clip}\left(F + \xi\frac{E-r}{8}; 0, 1\right). \tag{S9.4}$$

In this group, ρ is the daily recovery coefficient, μ the unmet-need damage coefficient, and ν the fatigue damage coefficient. E is exertion-equivalent executed hours: paid hours weighted by the selected intensity, plus care, training, travel and organization under their declared weights. r is actual rest, measured from the executed time ledger; the denominator is eight hours. ξ controls the daily fatigue response. Primes denote the next state.

These bounds apply to the model's health and fatigue variables. They are not permission to clip an overdrawn cash or inventory account. Likewise, a higher intensity changes the declared physical activity; it does not automatically multiply analytical labor value. Qualifications change through explicit training or other enabled events, not through the stationary education matrix alone.

GPU reading. Each person or household record can evaluate these elementwise operations in parallel after its delivered services and executed activities have been gathered. Shared household resources and personal hours must already have passed the transaction resolver. The kernel does not independently promise the same caregiver's time to two jobs.

14 Firms, markets, equipment and research effort

A firm plans from its observed sales, orders, supplier quotes and finances. Its proposed output is distinct from work in progress, completed stock, a sale and collected payment. Markets are interactions among identified buyers and sellers with limited search, actual offers and resource constraints. The quote rule below is one declared local policy within those interactions; it is not an equation that clears the whole economy. [M2, E4]

14.1 Forecast demand and propose starts

Using $\mathrm{L}(a, b; \lambda) = a + \lambda(b - a)$,

$$\widehat{d}' = \mathrm{L}(\widehat{d}, s + b; \lambda), \qquad q_{\mathrm{start}} = \left[\ell\widehat{d} - (z + j)\right]_+ . \tag{S9.5}$$

In this local notation, $\widehat{d}$ is the daily demand forecast; s and b are executed sales and distinct valid unfilled requests observed in the daily interval. The smoothing coefficient satisfies $0 \le \lambda \le 1$. Cancelled or repeated requests do not count as additional demand. z is saleable finished stock, j expected output already in work in progress, and ℓ is production lead time plus one day plus target buffer days. The product $\ell\widehat{d}$ and the stock terms are all output quantities.

For example, a forecast of 10 units/day, a four-day coverage target and 25 finished-plus-pipeline units propose 15 additional units. Inputs, staff, machine service, time and payment still determine whether those starts are accepted. The equation is not an order to manufacture missing resources. An updated forecast becomes visible only at its declared boundary; notation alone does not move an observation earlier. [M1, S9.2; M2, E4.1]

14.2 Separate machine condition from lifetime service

For one machine and one daily interval,

$$0 \le u \le \min\{R, q(c - d_m)\}, \qquad R' = R - u, \qquad q' = \mathrm{clip}(q - \delta u + \gamma m; 0, 1). \tag{S9.6}$$

Here R is remaining lifetime machine-service hours; $q \in [0, 1]$ is condition; c is nameplate service capacity for the day; and $d_m \in [0, c]$ is capacity occupied by maintenance downtime. u is executed productive machine service. Completed maintenance amount m restores condition at rate γ, while productive service reduces condition at rate δ. The coefficients carry the units needed for a dimensionless condition change.

Maintenance consumes its own materials, labor and downtime. It can occur even at zero condition because downtime uses nameplate capacity, but it never replenishes lifetime service R. Exhausted lifetime requires an actual replacement asset. Book depreciation, physical wear and replacement-cost valuation are separate records. [M2, E4.3]

For an enabled renewable natural stock, reuse neither the machine's units nor its interpretation. Define $s_R = R - u$ after actual extraction and use

$$R' = s_R \left[1 + \rho_R \left(1 - \frac{s_R}{K}\right)\right], \qquad 0 \le u \le R \le K, \quad K > 0, \quad 0 \le \rho_R \le 1. \tag{S9.7}$$

Now R, u, s_R, K are quantities of one natural resource, K is carrying capacity, and ρ_R its daily regeneration coefficient. Extraction precedes regeneration. Extracted goods and depleted natural stock have corresponding physical records; a regeneration formula does not grant access or supply extraction equipment. Nonrenewable stocks use zero regeneration. [M2, E8.2]

14.3 Update a seller's quote from admitted observations

$$p' = \mathrm{Quote}\left(p \exp\left[\eta_p \, \mathrm{clip}\left(\frac{b - x}{\max(\widehat{d}, 1_g)}; -1, 1\right)\right]\right). \tag{S9.8}$$

p is the seller's currency-per-unit quote, b its valid unfilled quantity, and x finished stock above its target, floored at zero. $\widehat{d}$ is the forecast quantity for the stated daily decision interval; 1_g is one unit of that output, in the same units. η_p is a dimensionless response coefficient. The capped ratio makes this a bounded response to the seller's own feedback.

Quote applies the existing admissible floor, upper cap and upward price-grid projection together. If that interval contains no grid point, the offer is unavailable. A quote supplies no inventory and forces no purchase. Wage quotes follow their own observations and rounding; they do not inherit product-price rounding. Analytical labor values do not automatically set either quote. [M1, S5.2, S9.3; M2, E4.2]

14.4 Research consumes effort before it can create mastery

Research separates exposure, technical mastery, legal permission and installed adoption. An observed publication can increase exposure without supplying trained staff or a license. A license can grant permission without creating competence. Installation still requires its own feasible work and equipment. [M2, E7]

Define the common hazard function $h(z) = 1 - \exp(-z)$ for $z \geq 0$. Then

$$h^* = \mathrm{L}(h_{\min}, h_0; (1 - xaq)^{\nu}), \tag{S9.9}$$

$$p_{\text{success}} = h(\Delta H / h^*), \qquad x' = \mathrm{L}(x, 1; h(\sigma)). \tag{S9.10}$$

Local symbol	Research meaning
x, a, q	Exposure, absorptive capacity and project relatedness, each in $[0, 1]$.
$h_0, h_{\min}$	Pioneer and minimum effort scales in effective research-hours, with $0 < h_{\min} \leq h_0$.
ν	Positive shape parameter controlling how useful exposure changes the effort scale.
ΔH	Actually executed effective research-hours in this daily slice.
σ	Nonnegative dimensionless sum of admitted signal strengths.

These a, q, x are research variables, not household health, machine condition or excess stock. Effective research-hours use declared project productivity coefficients, not the analytical labor-type conversion weights. With no useful exposure, $h^* = h_0$; useful exposure can lower the required scale toward $h_{\min}$. Patent restrictions affect permitted use separately.

One keyed outcome is drawn after each positive executed daily slice. Zero effort cannot succeed; unsuccessful effort remains in the resource history. Reusable knowledge does not cause past research labor to be charged again to every future copy. The GPU profile retains its versioned exponential/power tables, their rounding and their random-threshold contract. Sharing a mathematical function does not authorize a different numerical approximation. [M1, S5.2, S9.3]

15 Wages, taxes and bargaining

The wage mechanism joins a worker's needs and alternatives to an employer's funded plan. Keep four quantities distinct: the labor value of a customary reproduction bundle, its observed money cost, contracted gross compensation, and net cash actually received. Public services, household assets, care burdens, union organization and available alternatives affect these through different channels. None is replaced by an identity equating wages with labor values or productivity. [M2, E9.1]

15.1 Compute taxes once across a person's paid earnings

For thresholds $0 = b_0 < b_1 < \cdots < b_K$, marginal rates t_k, and nonnegative taxable income y,

$$T(y) = t_0 y + \sum_{k=1}^{K} (t_k - t_{k-1}) \, [y - b_k]_+ . \tag{S10.1}$$

Here T means tax, not available work time or a loan term. A marginal rate applies to income within its bracket. The last bracket has no upper threshold. On an additional gross payment g, income tax is $T(Y + g) - T(Y)$, where Y is the person's already paid taxable income in the period, shared across employers. A second job does not receive a duplicate unused allowance. The hinge form is an algebraic rewriting of the same bracket schedule. [M1, S10.1; M2, E20.2]

For a proportional employee charge τ_w, define

$$f(y) = (1 - \tau_w) y - T(y), \qquad N(w) = f(Y + w H_{\text{pay}}) - f(Y). \tag{S10.2}$$

w is the gross base rate in currency per ordinary hour. Actual work time $H > 0$ differs from premium-weighted payable time $H_{\text{pay}} > 0$: the latter sums each scheduled slice's hours times its pay multiplier. Gross compensation is $w H_{\text{pay}}$; $N(w)$ is incremental net compensation for this fixed schedule. Employee withholding partitions gross pay rather than adding another employer cost.

15.2 Find an acceptable wage interval

Let r_{net} be the worker's required net currency per actual work-hour. If every marginal net-pay slope satisfies $1 - \tau_w - t_k > 0$, then

$$g_* = f^{-1}(f(Y) + Hr_{\text{net}}) - Y. \tag{S10.3}$$

This is the additional gross payment needed to deliver the net requirement. With wage-grid spacing $\Delta w > 0$,

$$R = \lceil \max\{w_{\min}, g_*/H_{\text{pay}}\} \rceil_{\Delta w}, \qquad U = \left\lfloor \frac{M}{(1+\tau_e)H_{\text{pay}}} \right\rfloor_{\Delta w}, \tag{S10.4}$$

$$M = \text{clip}(\widehat{R} - \widehat{C} - Z; 0, B). \tag{S10.5}$$

The grid notation means $\lceil x \rceil_{\Delta w} = \Delta w \lceil x/\Delta w \rceil$ and analogously for downward rounding. R is the minimum acceptable gross rate and U the employer's maximum; $R > U$ means no feasible contract.

Local symbol	Meaning for this frozen candidate schedule
$w_{\min}$	Largest active technical, statutory and collective base-rate floor.
τ_e	Nonnegative proportional employer charge, additional to gross wages.
B	Funded, disjoint candidate allowance for total employer cost.
$\widehat{R}, \widehat{C}$	Expected realizable receipts and allocated nonlabor costs under admitted observations.
Z, M	Declared contingency/retained contribution and resulting payroll-cost ceiling.

Authorized research, maintenance or subsidized projects without immediate sales use $M = B$. A forecast limits willingness to pay but cannot fund current settlement. Candidate budgets come from one frozen plan and cannot each claim the same remaining money. [M2, E9.3]

Within the bracket containing the target net income z, inversion requires one affine calculation:

$$f^{-1}(z) = b_k + \frac{z - f(b_k)}{1 - \tau_w - t_k}, \qquad z \in [f(b_k), f(b_{k+1})]. \tag{S10.6}$$

The final bracket uses an infinite upper endpoint. Precompute $f(b_k)$ for the active law. Benefit cliffs or other nonmonotone total-income rules keep their existing finite search. Earlier payments and overtime commitments may change the inputs, so final acceptance rechecks the net-cash guarantee and shared resources. An unpaid fractional wage claim is not received net cash.

Worked tax and wage calculation: hypothetical teaching parameters

Take 10% income tax through 100 and 30% above it, plus a 5% employee charge. Prior paid earnings are $Y = 80$. A fixed schedule has $H = 4$ actual hours, $H_{\text{pay}} = 5$ payable hours and a required net rate of 6 per actual hour. Assume the legal floor is below the computed rate.

Then $f(80) = 68$, the target is $68 + 4 \times 6 = 92$, and $f(100) = 85$. Above 100 the net slope is 0.65, so total gross income must reach $100 + (92 - 85)/0.65 = 1440/13$. Additional gross pay is $g_* = 400/13$, giving base rate $80/13 \approx 6.15385$.

On an illustrative 0.01 wage grid, $R = 6.16$. If $M = 40$ and $\tau_e = 0.1$, then $U = \lfloor 40/(1.1 \times 5) \rfloor_{0.01} = 7.27$. At the minimum acceptable rate, gross pay is 30.80, incremental income tax is 5.24, employee charge is 1.54, and net pay is $24.02 \geq 24$. Employer cost is 33.88. This establishes a feasible interval; bargaining still selects an agreed rate. These are teaching inputs, not the source scenario's defaults.

15.3 Needs and the ability to refuse shape the worker's floor

For $n > 0$ willing available earning adults and normal day T_0, define the hourly household anchor $a_h = [C_{\text{cash}} - Y_h]_+ / (nT_0)$. Here C_{cash} is recurring cash cost after assured deliverable recurring in-kind provision; Y_h is funded regular nonwage income usable today. Already purchased provisions do not reduce this recurring flow cost. With refusal cover $D \geq 0$,

$$\phi(D) = \frac{3}{5} + \frac{2}{35} \min\left(D, \frac{21}{2}\right), \qquad r_{\text{net}} = \phi(D) a_h + c_w. \tag{S10.7}$$

D measures days of subsistence cover: unreserved household cash, usable necessities valued at the same quotes, and assigned funded refusal support not yet posted, divided by positive C_{cash}. Once a grant settles, it is in cash and leaves the unpaid-assignment component. Employment-conditioned benefits count only if they remain deliverable during refusal. c_w is this candidate's additional work cost per actual hour, excluding costs already counted in the normal bundle.

If $n = 0$ there is no wage application. If $C_{\text{cash}} = 0$, use the existing separate branch $\phi = 6/5$. The normal day remains the reference even when a job offers few hours: a satisfactory hourly rate can coexist with insufficient daily earnings. These conventions model bargaining constraints; they do not reduce physical needs when poverty weakens refusal. [M2, E9.2]

15.4 Bargaining is bounded and still requires consent

The source's local bargaining parameter, written β_{barg} here to distinguish it from occupational weights, is

$$\beta_{\text{barg}} = \frac{\max\{1, 7 + 5o + 4s + 3x - 4m - 3z\}}{20}. \tag{S10.8}$$

All five inputs lie in $[0, 1]$. o is the share of the frozen relevant workforce covered by an active collective mandate. s is the observed share in an opening active withdrawal, or a public conditional intention backed by the eligible person's one-day funded household-support assignment. x is the applicant's share of sampled qualified live alternative offers meeting R; without verified qualifying alternative evidence, set $x = 0$. m is the sum of squared effective wage-controller shares in observed accessible vacancy-hours. z is qualified willing replacement hours divided by requested hours, capped at one. Missing evidence and observed absence remain distinguishable. Empty workforce gives $o = s = 0$; no observed vacancies gives $m = 1$; absent replacement evidence gives $z = 0$. Zero requested hours permits no negotiation.

These are local material and organizational observations, not a universal class-power score. Common passive ownership does not merge independently controlled wage decisions; replacement hours are deduplicated by person-time. Supported intentions use one household grant even when several members consider action. [M2, E9.4]

For actual first offer q and submitted desired wage $Q \geq q$, round two uses

$$O_2 = \text{clip}\left(\lfloor \text{L}(q, Q; \beta_{\text{barg}}) \rfloor_{\Delta w}; R, U\right), \qquad R \leq U. \tag{S10.9}$$

The worker's claim Q is the greatest applicable claim among R; the base rate delivering the normal net anchor $a_h + c_w$ on this schedule; the original posted quote; a qualified live alternative adjusted to this job's costs and deductions; an effective collective target; and the prior agreed wage indexed by observed basket-cost change and the declared share of positive observed productivity growth. Productivity uses completed comparable output and paid-hour records, not a hidden future result. A missing or zero productivity comparator disables that adjustment; a missing or zero prior basket cost retains the prior nominal claim and records the status. Convert alternatives to the same ordinary gross base-rate units before comparison. Because the original posted quote enters Q while q is that quote capped at U, the active branch has $Q \geq q$. [M2, E9.3]

A claim is not a guaranteed raise. The source bounds search and bargaining, including two applications and two offer rounds. Individual acceptance, confirmed withdrawals and final joint feasibility decide whether work occurs. A failed negotiation does not force market clearing or compulsory work.

16 Credit, investment and collective action

Finance connects production to the dates on which money returns. Collective organizations connect individual decisions to shared support and authority. Both use actual funds and claims: a treasury, a loan approval, a receivable and a paid transfer are different objects. This section's repeated letters are local; in particular F below is a treasury, not fatigue or the analytical research-labor matrix.

16.1 A bank must satisfy every lending limit

For the source's secured-business-credit policy,

$$L_{\text{new}} = \left[\min \left\{ L_{\text{req}}, \frac{T(\theta \widehat{F} - d)}{1 + rT}, \lambda V_c - L_{\text{sec}}, \frac{E - k_{\min}\text{RWA}}{k_{\min}\omega}, L_{\text{liq}} \right\} \right]_+ . \tag{S11.1}$$

Term	Meaning and restriction
L_{req}	Nonnegative requested principal.
$T, r, \widehat{F}, d, \theta$	Positive daily installment count; nonnegative daily rate; observed daily cash operating surplus; existing daily debt service; servicing fraction in $[0,1]$.
$\lambda V_c - L_{\text{sec}}$	Appraised collateral V_c in currency, times advance fraction $\lambda \in [0,1]$, less existing secured commitments L_{sec} in the same currency against that collateral.
E, RWA	Bank equity and existing risk-weighted assets.
$k_{\min}, \omega$	Positive minimum capital ratio and new-loan risk weight.
$L_{\text{liq}} \geq 0$	Unreserved first-use liquidity in currency from reserves and an already funded undrawn credit line.

The advance fraction satisfies $\lambda \in [0,1]$; d, V_c, L_{sec} and RWA are nonnegative. Observed surplus $\widehat{F}$ and equity E may be negative, in which case the affected cap blocks new lending. The positive part bounds a new decision; it does not erase existing debt or losses.

The caps respectively limit the request, payment burden, collateral, bank capital and liquidity. Straight-line principal makes $L(1/T + r)$ the largest installment; a different amortization rule needs its full schedule. Each accepted approval updates shared collateral, capital and liquidity reservations before the next conflict. An unsecured-credit variant must supply another explicit exposure rule. [M2, E5.2; E17]

Loan origination creates matched bank loan/deposit entries and borrower deposit/debt. Actual cross-bank payment then requires reserves. A bank may have positive equity but lack settlement liquidity; loan losses may impair equity even when some reserves remain. A central-bank advance adds both reserves and bank debt, not free equity. Impairment changes creditor valuation without automatically cancelling the debtor's contractual face or deleting deposits. These distinctions allow credit and settlement problems to propagate through actual claims. [M2, E17.4]

16.2 Find the largest cash shortage along the plan

For ordered cash events i, let $c_i = \widehat{C}_i - \widehat{R}_i$ and compute

$$P_0 = 0, \qquad P_k = \sum_{i=1}^{k} c_i, \qquad B_{\text{req}} = \max_{0 \leq k \leq n} (b_k + P_k), \qquad G_{\text{finance}} = [B_{\text{req}} - D_0]_+ . \tag{S11.2}$$

$\widehat{C}_i$ and $\widehat{R}_i$ are nonnegative planned outflows and expected receipts in one currency. Order is by day, phase and persistent identifier. $b_k \geq 0$ is the required continuity cushion at that prefix, and D_0 actual total opening cash, including cash already earmarked for payments inside this plan. B_{req} is an opening-balance test, not another cash debit. The empty prefix matters. [M2, E22.4]

The same eventual margin can require different finance

An outflow of 8 followed by a receipt of 10 gives cumulative net costs $(0, 8, -2)$. With no cushion, the plan needs 8 initially despite its eventual surplus of 2. A constant cushion of 3 raises that need to 11. Actual execution still requires the receipt to settle before spending it; the forecast alone supplies no money.

For replacement, earmark only actually available cash:

$$\Delta B_{\text{rep}} = \text{clip}(K_{\text{target}} - B_{\text{rep}}; 0, D_{\text{free}}). \tag{S11.3}$$

K_{target} is observed delivered/installed replacement-basket cost, B_{rep} the existing earmark inside the same cash balance, and $D_{\text{free}} \geq 0$ settled unreserved cash after prior obligations and the continuity cushion. This changes availability, not total cash or physical capacity. Depreciation booked against unsold goods is not money recovered for replacement. [M2, E22.5]

The retained up-front rent bid is $\nu\,\text{clip}(V_{\text{pre rent}} - Z; 0, B)$: $V_{\text{pre rent}}$ is estimated value before rent, $Z \geq 0$ retained-value requirement, $B \geq 0$ rent allowance and $\nu \in [0, 1]$ bid fraction. Access remains restricted to the actual lease interval. Investment and claim comparisons reuse discount weights $d_s = (1 + r_d)^{-s}$ over a finite horizon, where s is the number of elapsed daily periods and r_d is the scenario's permitted daily discount rate under the source convention. The mathematical domain $r_d > -1$ does not itself authorize a different rate choice. For the applicable dated net cash-flow changes, the total is the dot product $d^{\mathsf{T}}\Delta\widehat{\text{CF}} = \sum_s d_s \Delta\widehat{\text{CF}}_s$. Each comparison retains its own eligible flows and initial-cost convention; a claim valuation uses its own eligible payment vector. A terminal receipt enters once under its declared rule. A positive forecast neither installs a machine nor creates a buyer. [M2, E22.7–E22.9]

16.3 Support households once, even when several members act

Let $a_h \geq 0$ be a distinct eligible household's capped residual support need after valid prior reservations, and $F \geq 0$ uncommitted organizational cash. Then

$$A = \sum_h a_h, \qquad \lambda_F = \begin{cases} \min(1, F/A), & A > 0, \\ 0, & A = 0, \end{cases} \qquad b_h = \lambda_F a_h. \tag{S11.4}$$

Compute the residual need from still-unprovided minimum services, current unreserved cash and other funded unsettled reservations. Usable provisions and delivered aid already reduced service needs; settled cash aid is already in cash. The letter a_h here is a grant capacity in currency, not the hourly wage anchor. With already validated eligible household capacities $(2, 4, 6)$ and treasury 6, reservations are $(1, 2, 3)$, irrespective of how many candidate adults belong to each household. This scaling establishes neither eligibility nor consent. [M2, E10.2]

The grant commits once if at least one eligible individual confirms. Every eligible action that independently confirms can join the bundle, subject to shared time and resource checks. If nobody confirms, release the reservation without same-day redistribution. A vote or another household member's decision cannot compel participation. Integer payments retain the numerical profile's exact rounding/claim rules; the proportional identity does not itself specify allocation of indivisible remainder units.

16.4 Tentative willingness is different from an executed strike

Beliefs use the shared affine update with admitted observations; absent observation leaves a belief unchanged. For a supported authorized or self-initiated dispute, the source's illustrative willingness rule is

$$p_i = \mathrm{sigmoid}(2g + s + u + e + n - 2c - r - 3/2), \qquad \mathrm{sigmoid}(x) = \frac{1}{1+\exp(-x)}. \tag{S11.5}$$

Input	Person's locally observed or learned feature, all in $[0,1]$
g	Grievance from compensation, pay-delivery, hours and condition shortfalls against declared targets.
s, u, e	Solidarity, trust in this organization, and perceived collective efficacy.
n	Observed contacted-peer support share; no reports gives zero.
c	Anticipated forgone pay divided by seven days of household reproduction cost, clipped to $[0,1]$.
r	Perceived sanction risk from admitted observed outcomes.

For the cost denominator zero case, use $c = 0$ if forgone pay is zero and $c = 1$ otherwise. Missing prices retain the permitted last estimate with its stale status. The score lies in $[-4.5, 4.5]$, allowing a finite pinned table. One keyed draw creates tentative willingness; actual withholding additionally requires consent, time, support and material eligibility. These coefficients are illustrative assumptions, not a universal class-consciousness measure or probability of revolution. [M2, E10.1–E10.3]

16.5 Institutions and crises act through the same constrained events

State policies can change taxes, wage floors, collective rights, public provision, credit facilities or ownership. Each requires a named authority, valid procedure, effective date and the resources needed for administration and delivery. Proposal, appropriation, performed administration, legal activation and delivered service are separate stages. The accumulation-led default and Keynesian variant change explicit priorities; an ideology label supplies no goods, money or worker control. Revolutionary transformation likewise requires implemented authority and property/control changes. [M2, E20–E21]

Government and person-level policy scores reuse weighted sums but have different inputs, preferences, thresholds and powers. Invalid required forecasts make a choice unavailable; a high score only selects a proposal. Public provision may lower private cash expenses while strengthening refusal capacity and changing available hours, so its net money-wage effect is resolved through the actual mechanisms.

Delayed collections, unmet orders and unpaid wages can reduce later spending and production. Loan losses can restrict bank capital; failed withdrawals can expose liquidity shortages; reduced financing can interrupt payroll and investment. These feedbacks can support an endogenous contraction when their modules are enabled. A "crisis" label remains a report, not a hidden command that creates a shock or forces recovery. The simplified formulas preserve these channels without claiming that a calibrated historical crisis has been reproduced. [M2, E19.5; E22.9; E23]

GPU reading. Local policy scores and caps are parallel elementwise work. Household totals and account legs use grouped reductions; financing uses an ordered prefix scan followed by a maximum. All proposed payments, grants, loans and jobs still pass the complete-event resolver before changing shared state. These mathematical forms do not relax the original observation, rounding or authority boundaries.

17 How the complete model runs on a GPU

17.1 Separate proposal work from contested commitments

At each permitted observation boundary, parallel policy kernels read their permitted snapshots. A snapshot is a declared view of the economic state at that point in time. It need not reveal every agent's private information, and it must not reveal future state.

Each proposed event includes its persistent ID and priority, parties and consent, authority, protected object versions, gross resource legs, partial-fill rule, and every mutable state item it reads or writes. This complete dependency list is its *footprint*. For example, a purchase may touch inventory lots, two deposits and bank reserves; a hiring event may additionally touch personal time, overtime records, employer allowance and a household support reservation.

Distinguish a protected contract or title version from resource availability. A contract change may invalidate an event, while availability should be rechecked after earlier events settle. Include all permitted alternative inventory lots and shared collateral or eligibility predicates in a conservative footprint. Omitting a shared dependency can make apparently independent proposals conflict.

17.2 A deterministic dependency schedule

For each mutable resource, link consecutive touching events in the source's canonical priority order. A canonical order is an order fixed by the model, such as economic priority followed by a persistent-ID tie-breaker. All dependency edges point forward in that same order, so the combined graph is acyclic.

Execute the *ready frontier*: events whose predecessors have finished. Its events have disjoint declared mutable footprints, so they can validate and commit in parallel. Rejected events release their successors too. Publish accepted changes before checking dependent events. If an undeclared mutable access is discovered, stop before the affected commit and rebuild the remaining schedule.

Why this preserves the serial result is a conditional argument. With complete footprints, identical event payloads and handlers, and the same arithmetic, independent events commute: swapping their order changes neither result. Repeatedly exchanging independent neighboring events transforms the canonical serial sequence into the frontier execution order. Conflicting events are never swapped. Adaptive observations must still occur at their original subphase boundaries; this argument cannot justify precomputing a proposal from information it would not yet know. [M1, S12.1; M2, G6–G7]

The failed-buyer example in Section 5 is one consequence. Rejecting the earlier event completes that dependency step and exposes the remaining stock to its successor. A hot reserve account or heavily contested input can create a long chain. GPU execution preserves that dependency rather than awarding resources to whichever thread runs first.

17.3 The narrower prefix-allocation shortcut

There is a faster formula for one divisible pool of capacity B, ordered prevalidated demands a_i, disjoint private constraints and no intervening inflows:

$$f_i = \min\left(a_i, \left[B - \sum_{j<i} a_j\right]_+\right). \tag{17.1}$$

f_i is the partial fill. An exclusive prefix scan gives the sum of earlier demands. For $B = 5$ and demands $(2, 4, 1)$, fills are $(2, 3, 0)$.

The conditions are essential. With *indivisible* jobs demanding $(6, 4)$ from capacity 5, the first can fail and the second succeed. The prefix of proposed demands would incorrectly block the second. Likewise, a multi-resource job can fail for lack of cash while its material allocation ought to remain available. Use the complete resolver unless the shortcut's narrower equivalence has been established. [M1, S12.1]

17.4 Data organization and host responsibilities

The large evolving state uses *structure of arrays*: separate arrays for fields such as person ID, health, fatigue, household link and available hours. Sparse edge tables store employment, ownership, loans, communication and recipe dependencies. Sorted edges or compressed sparse rows group relationships by owner or source. Permanent economic IDs remain distinct from dense storage slots.

The design keeps large state and intermediate buffers on the device between kernels, and batches necessary transfers. These choices are consistent with NVIDIA's published guidance on data residency, transfer costs and coherent memory access [R1]. They are design targets, not measured speedup claims for this simulator.

A Rust host can manage configuration, commands, durable checkpoints, output and rare structural work. The inherited concrete backend is CUDA, while the equations and event contract are backend-independent. An alternative runtime must supply the required integer widths or exact fallbacks, sparse and graph operations, synchronization and replay behavior. Choosing a tensor library does not automatically implement the transaction scheduler or legal/contract handlers.

Work	**Repeated operation**	**When cached inputs become stale**
Household and person policies	Elementwise arithmetic; gathers of actual services and hours.	New observations, delivered services or executed activities.
Recipe packets	Checked integer multiply, divide and minimum.	Recipe, units, stage timing or linear-invoice assumptions change.
Taxes and wage bounds	Threshold lookup, positive parts, affine inversion where valid.	Law, paid-income record, schedule or employer allowance changes.
Goods and occupational values	Four sparse block products or another verified solve.	Any selected technique, reproduction/training coefficient or numerical context changes.
Research and plan totals	Two dot products after values are cached.	The analytical context or research requirement catalogue changes.
Knowledge support	Neighbor gathers and set union/bitwise OR.	Requirements, access or other relevant graph versions change.
Financing plans	Ordered prefix scan followed by a maximum.	Cash-flow timing, forecasts, cushions or opening cash changes.
Event acceptance and posting	Stable ordering, dependency frontiers, exact leg reductions.	New event batch, changed shared state or invalidated object version.

Sparse arithmetic and graph operations are different workloads. A dense matrix over every pair of agents is unnecessary. A sparse linear-algebra library provides selected building blocks, not the entire simulation. cuSPARSE, for example, documents sparse vector, matrix-vector and matrix-matrix operations [R2]. Actual support, formats, algorithms and performance must be checked for the selected backend.

Host interventions that affect the economy become versioned events at declared boundaries. Memory compaction can change physical slots without changing logical identities. If an event buffer fills, pause before the affected commit and recover by the declared resize/spill policy; buffer exhaustion must not become a fictional shortage, failed withdrawal or political outcome.

17.5 Profiles and the meaning of replay

A *profile* fixes how the same model vocabulary is represented and executed. It is part of a run's identity.

Profile	Contract	What remains to demonstrate
REF-5.1	General source rational accounts, Decimal50 policies, keyed draws and ordered events.	Complete original-profile implementation and replay.
GPU-D1	Bounded exact integer accounts, coherent packets, explicit residuals, versioned integer policies and retained priority/observation boundaries.	Same-profile CPU/GPU agreement and portable device replay.
GPU-F1	Same account and event structure, with selected separately versioned FP64 policies/analytics.	Same-stack reproducibility and measured decision differences; cross-device bits are not implied.
Reduced P0	Inherited Python implementation with its own integer units and narrower economic scope.	A completed GPU port or an explicit numerical migration.

FP64 means 64-bit floating-point arithmetic. Even when a floating-point diagnostic is accurate within a tolerance, a tiny difference near a decision threshold can change an action. Analyst-only diagnostics can have a tolerance contract; once an agent uses a result, its decision-relevant computation must satisfy the selected profile.

Pin algorithms, versions and operation ordering. cuSPARSE documents reproducibility limits, including no such guarantee for its transpose operations [R2]. Explicit transposed storage may permit a chosen non-transpose path, but does not by itself certify the complete run. CCCL distinguishes repeatability on one GPU from supported cross-GPU guarantees, with version and operation restrictions [R3]. Whole-simulator replay needs implementation-level evidence.

A checkpoint includes pending obligations, commitments, queues, object and graph versions, policy-table versions, observation history and random-key context. A seed plus current visible balances is insufficient. Independent complete worlds can be batched for experiments; distributing one interdependent economy introduces additional synchronization requirements. [M1, S13]

18 What is specified, what was checked, and what remains

18.1 The boundary of this editorial revision

This edition expands the explanations of v6.1. It adds an economic orientation, local definitions and units, worked examples, a notation primer, a glossary and implementation context. It does not add a new behavioral rule, tax schedule, research convention, numerical profile or backend claim. Equation tags beginning with S in the policy chapters intentionally retain v6.1's labels; the cross-reference appendix maps the rest.

The examples use deliberately small hypothetical numbers. They explain arithmetic and causality. They are not scenario initializers, empirical parameter estimates, complete economies or examples of historical calibration.

The main text explains the meaning and use of the simplified formulas without requiring prior conversation. An implementer still needs the selected scenario's full catalogues, initial state, parameter domains, institutional handlers and failure rules. Those exhaustive records remain in the retained reference. A readable formula guide is not a substitute for missing implementation code.

18.2 Evidence attached to this edition

Evidence	What it supports	What it does not establish
New example checks	Exact arithmetic for the illustrative transfers, packets, claims, tax/wage example, values, research totals, residual bound and allocation examples.	A complete economic transition engine or device execution.
Retained v6.1 checks	The prior report records 10 families and 31,973 exact-integer/rational assertions comparing formula reductions.	31,973 independent economies; large-scale convergence or performance.
Retained v6.0 checks	Selected earlier scheduler, allocation, resource and sparse-algebra checks under their stated scope.	A full GPU implementation or unqualified cross-profile equivalence.
Reduced P0 source/results	The inherited implementation and its explicitly limited scenario coverage.	Implementation of every module described in the wider specification.
Document checks	Rendered layout, readable equations/tables, internal navigation and preserved attachment bytes.	Empirical validity of the economic hypotheses.

The new example-check script and its results are attached separately. Earlier test results retain their original attribution; they are not relabeled as new v6.2 runs. The exact source/build files for this explained edition are included in an attachment for future editing.

18.3 From specification to a runnable model

For an enabled scenario, supply an initializer, observations, policy, transition handler, parameter domain and failure path for every module. Then compare a same-profile CPU oracle and GPU execution at each named subphase, including accepted event IDs, balances, reservations, claims, time use and queues.

Concrete remaining risks include integer overflow in intermediates, unrepresentable packets, incomplete event footprints, stale context caches, incorrect rounding and information becoming visible too early. Test those implementation risks when the engine exists. Exercise restart and compaction against the same logical history, and measure full simulated days on declared hardware.

Economic validation is a separate task: choose data or stylized observations, identify measurable outputs, calibrate and compare mechanisms, and disclose sensitivity to behavioral assumptions. Neither the sparse equations nor the resource scheduler guarantee a falling profit rate, revolution, depression or recovery. Those remain outcomes to investigate under an explicit scenario.

A A short reader's glossary

Term	Meaning
Accrual	Recognition of an earned obligation, which can precede its actual cash settlement.
Adoption	Installing or using a method in production after knowledge, access and resource conditions are met.
Authority	A valid institutional power to perform an action; separate from physical ability to seize or use an asset.
Canonical order	The model's fixed priority and tie-breaking order, independent of GPU scheduling.
Closure	An additional relationship that makes an analytical system determinate; or, for sets, repeatedly adding all required dependencies. The local context specifies which.
Conservation	An invariant for the particular instrument or physical transfer being checked. It is not a claim that production leaves every material unchanged.
Context version	A specific selection of techniques, units, classifications and other analytical inputs. Cached outputs are valid only for the matching version.
Counterparty	The other party to a contract or claim. Both sides refer to the same underlying obligation.
Event bundle	All legs and conditions of one indivisible accepted economic action.
Execution barrier	A point where required preceding computations are complete before a subsequent stage can observe or change the state.
Footprint	All mutable records and predicates that an event can read or write, including shared constraints.
Gross inputs / output	Inputs required before returns; or total output including intermediate use. These differ from net use and final delivery.
Impairment	A creditor's reduction in the recognized value of an asset under the accounting rules; not automatic cancellation of its legal face.
Liquidity	Availability of the relevant asset to settle a payment when due.
Mastery	An actor's technical competence to use knowledge. It is distinct from discovery, permission and installation.
Observation budget	A limit on what an agent can inspect during a decision, not a cap on the underlying relationship network.
Packet	A compiled whole number of activity units whose linked resource legs are representable exactly.
Persistent ID	An object's lasting logical identity, independent of its current memory location.
Prefix scan	All running totals of an ordered list; an exclusive scan omits the current entry.
Reference context	A declared analytical production/reproduction setting used to calculate comparable requirements.

Term	Meaning
Reservation	A commitment of an actual resource that prevents conflicting use; it is not necessarily its final consumption or payment.
Residual	The mismatch of a numerical candidate with its equation; in the occupational closure, a separately defined common remainder κ. These meanings are distinct.
Settlement	Completion of the prescribed transfer legs, making an admitted payment actually received.
Snapshot	The permitted view of state at a named observation boundary.
Solvency	The relation of valued assets to liabilities and equity under the selected conventions.
Sparse	Stored through represented nonzero entries or relationships, rather than all possible pairs.
Tranche	A separately tracked portion of a claim with compatible rate, dates, denomination and accrual rules.

B Find the formula you need

Reader's question	This edition	Compact source
What is simulated?	Sections 1–3: agents, categories, actual history and daily order.	S1, S12; v6.0 G1, G8, G12–G15.
How do amounts and payments change?	Section 5: typed stocks and complete events.	S2; v6.0 G3, G6.
How do exact resource ratios survive integer storage?	Section 6: packets, gross inputs and obligations.	S3–S4.
How are bounds, forecasts and random choices evaluated?	Section 7: grids, blending, tables and keyed draws.	S5.
How are ordinary labor requirements calculated?	Section 8: recipe orientation and sparse solves.	S6.1; equation S6.5.
How are labor types compared?	Section 9: optional reproduction closure and normalization.	S6.2, equations S6.2–S6.5.
How is research counted?	Section 10: attempts, plan totals, gross outputs and history.	S6.3, equations S6.6–S6.7.
When is a numerical answer reliable?	Section 11: contraction certificate and residual bound.	S7.
What knowledge must be available?	Section 12: set closure and readiness.	S8.
How do needs and activity change people?	Section 13: deficits, one clock, health and fatigue.	S9.1.

Reader's question	This edition	Compact source
How do firms, prices, machines and research respond?	Section 14: local operating policies.	S9.2–S9.3.
How are wage bounds and bargaining derived?	Section 15: taxes, refusal cover and negotiation.	S10.
How do finance, support and institutions enter?	Section 16: credit caps, dated finance and collective choice.	S11; v5.1 E20–E22.
What actually runs in parallel?	Section 17: dependency frontiers, cache boundaries and profiles.	S12–S13.
What has been verified?	Section 18: new examples and inherited evidence.	S14–S15.

Precedence. This edition explains v6.1 and the retained source rules. It does not supersede a scenario's explicit numerical contract or invent a new initializer. The v6.1 algebraic presentations continue to govern their corresponding reductions; v6.0's G-sections govern selected GPU execution changes; the preserved v5.1 E-sections and Foundation govern the remaining economic definitions. The new examples illustrate those rules and are not additional default parameters.

C Sources and preserved reference material

C.1 Model sources

M1 *Unified Economic Simulator v6.1: Simplified GPU Formulas*, 27 October 2026. The 15-page source of this explanatory revision, retained as `prior_v6_1_formula_manual.pdf`. References such as S6 or S10 identify its section or equation labels.

M2 *Unified Economic Simulator v6.0: GPU Execution Specification*, with the complete retained v5.1 specification and Foundation, 27 October 2026. The 196-page reference remains attached as `prior_v6_0_complete_reference.pdf`. G labels refer to its GPU revision; E labels and Foundation refer to its retained economic source.

M3 The inherited source package, specification data, audit/coverage records, and versioned verification scripts/results embedded in M1 and M2. Their limitations remain part of the evidence.

The economic explanations in this edition describe the proposed model and its inherited assumptions. Model citations establish that provenance; they do not establish empirical accuracy or scholarly consensus.

C.2 Primary implementation references

These references were checked on 27 October 2026. The relevant guidance supports particular implementation considerations, not the complete simulator's correctness or performance.

R1 NVIDIA, *CUDA C++ Best Practices Guide*, especially data transfer between host and device, keeping intermediates resident, and coherent memory access.
`https://docs.nvidia.com/cuda/cuda-c-best-practices-guide/index.html`

R2 NVIDIA, *cuSPARSE Documentation*, introduction, sparse operations and section 2.4, Result Reproducibility.
https://docs.nvidia.com/cuda/cusparse/index.html

R3 NVIDIA, *CUDA Core Compute Libraries: Determinism*. Operation, configuration and fixed-version restrictions apply. This is the moving unstable documentation branch.
https://nvidia.github.io/cccl/unstable/cccl/determinism.html

C.3 What is inside this PDF

The complete v6.1 input and every inherited attachment are preserved. They include the complete prior reference, the reduced P0 source/results, specification and mechanism data, previous checks and editable v6.1 sources. New files contain this edition's editable LaTeX/build sources, arithmetic examples and their results. The attachment manifest records names, byte sizes and SHA-256 hashes.

Use a PDF reader with an attachments panel to extract these files. Some browser and phone viewers hide embedded attachments; the main explained text does not require them. Older source-prefixed content-credential data describes its original artifact and does not attest this revision.

New material:

- `v6_2_explained_sources.zip`: editable text, build/packaging code and example checks.
- `v6_2_example_checks.py` and `v6_2_example_results.json`: executed teaching-fixture arithmetic and its report.
- `v6_2_attachment_manifest.json`: provenance and byte-preservation record.